Innovation Yes, real!

CampusShield

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KNOWLEDGE IS POWER

Estd : 1984

COURSE : 24ITL34 – DESIGN THINKING LAB

EXERCISE : 2

DATE : 3.9.2026

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T2 : STEEP Trends Analysis

STEEP ANALYSIS

SOCIAL & DEMOGRAPHICS

- Improves student safety awareness
- Easy emergency reporting
- Quick communication with security
- Supports students, staff and faculty
- Creates a safer campus environment

TECHNOLOGY

- GPS-based real-time location sharing
- One-tap SOS emergency alert
- Mobile app-based safety system
- Real-time notifications
- Secure user data and authentication

ECONOMY

- Reduces campus security costs
- Affordable software-based solution
- Minimizes emergency response delays
- Reduces dependence on physical infrastructure
- Supports efficient use of security resources

ENVIRONMENT & NATURE

- Reduces paper-based reporting
- Promotes digital communication
- Helps during natural emergencies
- Supports quick location-based assistance
- Encourages a safer campus environment

YOUR DESIGN CHALLENGE ISSUE

"How might we create a mobile application that provides students with quick, reliable and secure emergency assistance on campus?"

POLITICS & LEGAL

- Protects student privacy
- Secure handling of personal data
- Follows campus safety policies
- Requires user consent for location sharing
- Supports emergency reporting procedures

T3 : STEEP Analysis Matrix

STEEP Analysis Matrix

High Impact

HIGH IMPACT – UNCERTAIN TO OCCUR

- Network failure during emergencies
- GPS unavailable in certain areas
- High number of false emergency alerts
- Cybersecurity threats
- Low user adoption

HIGH IMPACT – LIKELY TO OCCUR

- One-tap SOS emergency alert
- GPS-based real-time location sharing
- Real-time security notifications
- Quick communication with campus security
- Secure student data protection

Uncertain to Occur — Likely to Occur

LOW IMPACT – UNCERTAIN TO OCCUR

- Temporary server downtime
- Device compatibility issues
- Occasional notification delays
- Changes in campus safety policies
- Limited usage during non-emergency situations

LOW IMPACT – LIKELY TO OCCUR

- Regular app notifications
- Routine security reporting
- Minor app updates
- Normal location tracking
- User feedback and suggestions

Low Impact

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T4 : STEEP Analysis Prioritization

STEEP Analysis Prioritization

Discuss 2 key trends from the High Impact-Likely to Occur quadrant:

1. **GPS-based real-time location sharing**
   Helps security teams quickly identify the student's location during an emergency.
2. **One-tap SOS emergency alert**
   Allows students to send an instant emergency alert to campus security with minimal effort.

Discuss 2 trends from the High Impact-Uncertain to Occur quadrant:

1. **Network failure during emergencies**
   Poor or unavailable internet connectivity may affect SOS alerts and real-time location sharing.
2. **Cybersecurity threats**
   Unauthorized access or data breaches could affect the privacy and security of student information.

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T5 : Strategic Priorities Matrix

STRATEGIC PRIORITIES MATRIX

	URGENT	LESS URGENT
IMPORTANT	1. One-tap SOS emergency alert 2. GPS-based real-time location sharing 3. Instant notification to campus security 4. Live emergency location tracking 5. Quick communication with security team 6. Immediate emergency response	1. Secure user data & authentication 2. Real-time security notifications 3. Security dashboard for administrators 4. Secure storage of incident records 5. User privacy and consent management 6. Regular system maintenance and updates
LESS IMPORTANT	1. Quick emergency reporting 2. False-alert verification 3. Incident status updates 4. Reporting unsafe campus locations 5. Emergency communication support 6. Alert acknowledgement by security staff	1. Digital/paperless reporting 2. User feedback and suggestions 3. App interface improvements 4. Safety awareness tips 5. Periodic performance reports 6. Additional app features

T6 : Synthesis (Sense Making)

SYNTHESIS: MAKING SENSE OF STEEP ANALYSIS AND STRATEGIC PRIORITIES

Assessment Questions	Synthesis: Sense Making
What relationships among the trends do you perceive? How are they related? Why are these relationships important?	SOS alerts, GPS tracking and real-time notifications are connected to provide faster emergency response and improve campus safety.
What opportunities and/or challenges need immediate attention going forward for your design challenge? And why?	Opportunities include faster emergency response and better communication. Challenges include network issues, false alerts and cybersecurity.
What would it take to create positive change on this issue relating to your design challenge?	A simple safety app with reliable SOS, GPS, notifications and secure data management is needed to improve campus safety.
Who else would be interested in this issue? Why should they care? What conversations would you have with them?	Students, security staff, faculty, management and parents are interested because they want a safer campus and faster emergency response.

T7 : Activity System Mapping

MAPPING ORGANIZATION ACTIVITY SYSTEM

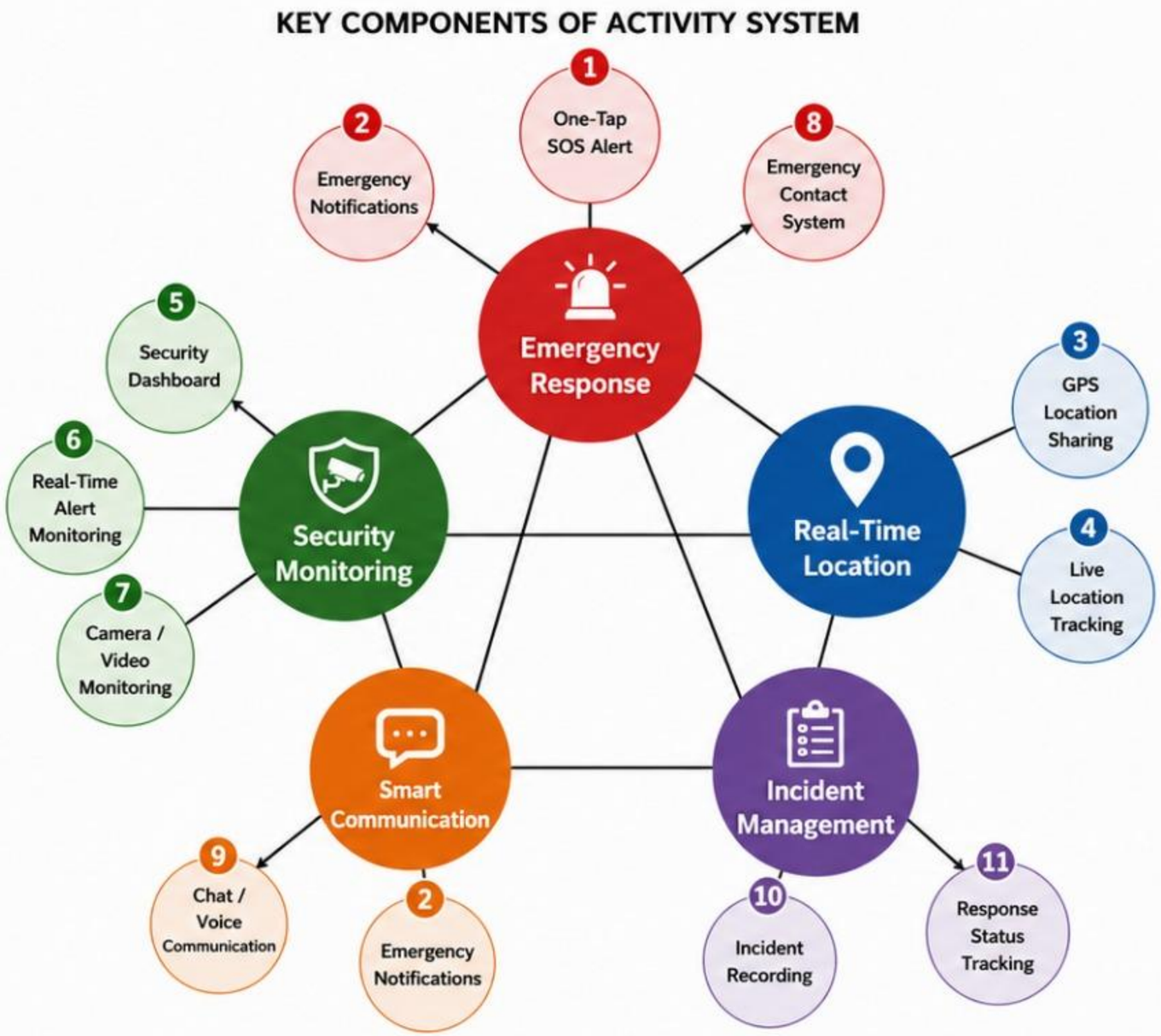
OBJECTIVES / MISSION STATEMENT

To provide a fast, reliable and secure campus safety system for students through real-time emergency alerts and location sharing.

Strategy #1 Emergency Response	*Strategy #2 Real-Time Location*	*Strategy #3 Security Monitoring*	*Strategy # 4 Data & User Security*
SOS Alert	GPS Tracking	Security Dashboard	User Authentication
Emergency Notification	Live Location	Incident Monitoring	Data Encryption
Quick Response	Safe Route	Incident Records	Access Control
Alert Tracking	Location History	False Alarm Detection	Security Alerts

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T8 : Key Components of Activity System



STRATEGIC HUBS

- Emergency Response
- Security Monitoring
- Real-Time Location
- Smart Communication
- Incident Management

KEY COMPONENTS – 11

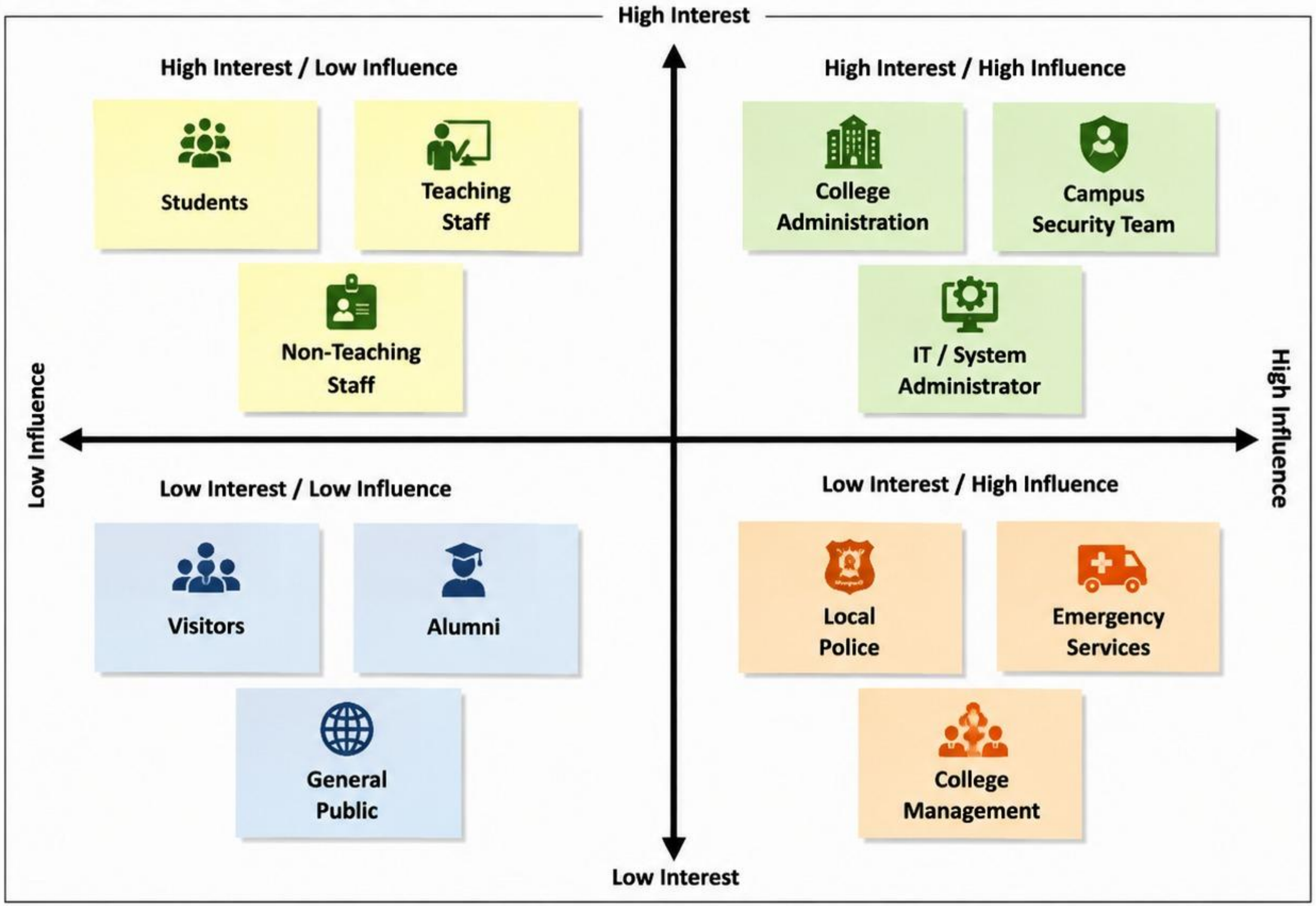
1. One-Tap SOS Alert
2. Emergency Notifications
3. GPS Location Sharing
4. Live Location Tracking
5. Security Dashboard
6. Real-Time Alert Monitoring
7. Camera / Video Monitoring
8. Emergency Contact System
9. Chat / Voice Communication
10. Incident Recording
11. Response Status Tracking

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T9 : Stakeholder Mapping Matrix

STAKEHOLDER MAPPING MATRIX

(Identify your relevant key stakeholders)

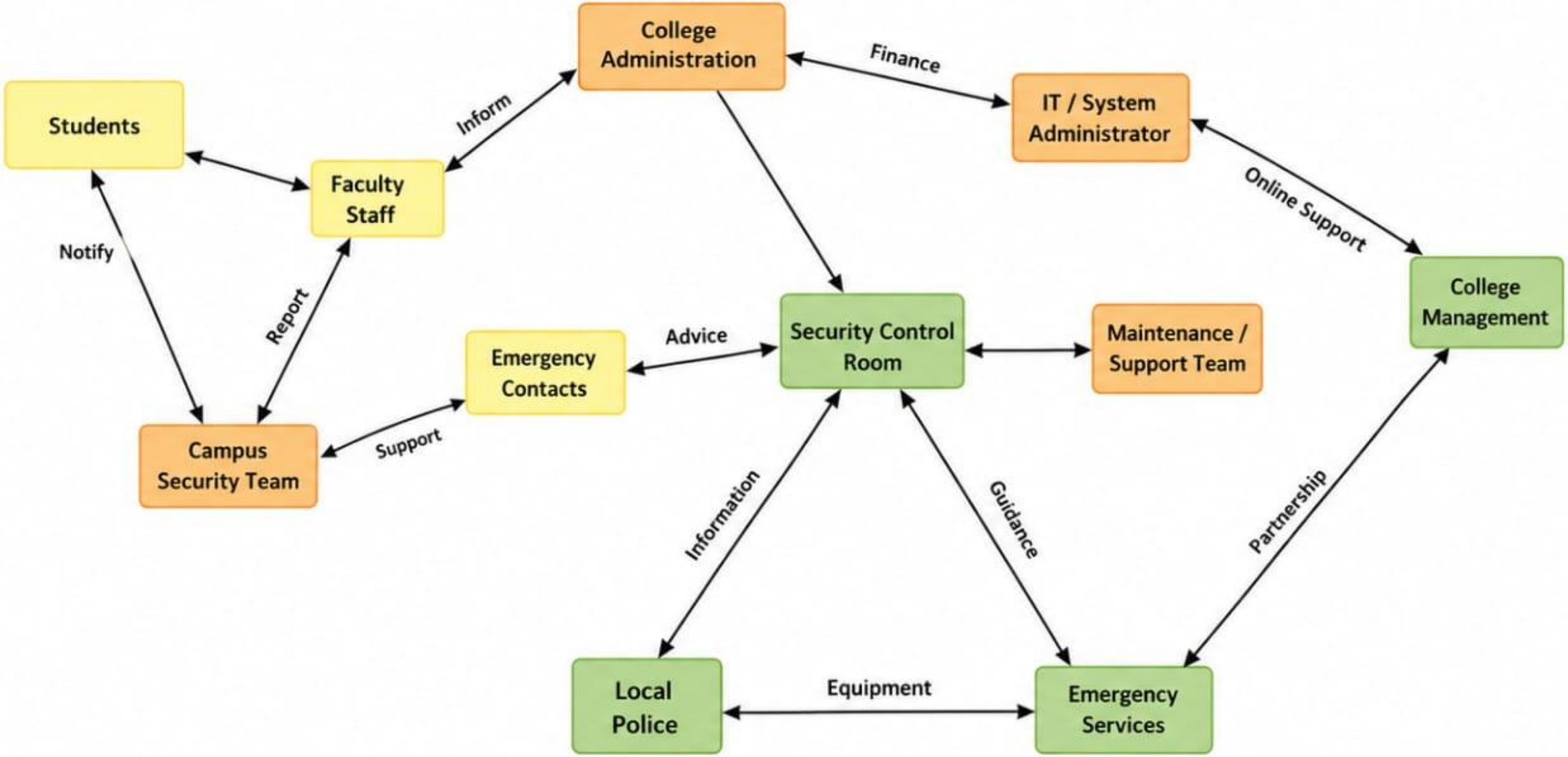


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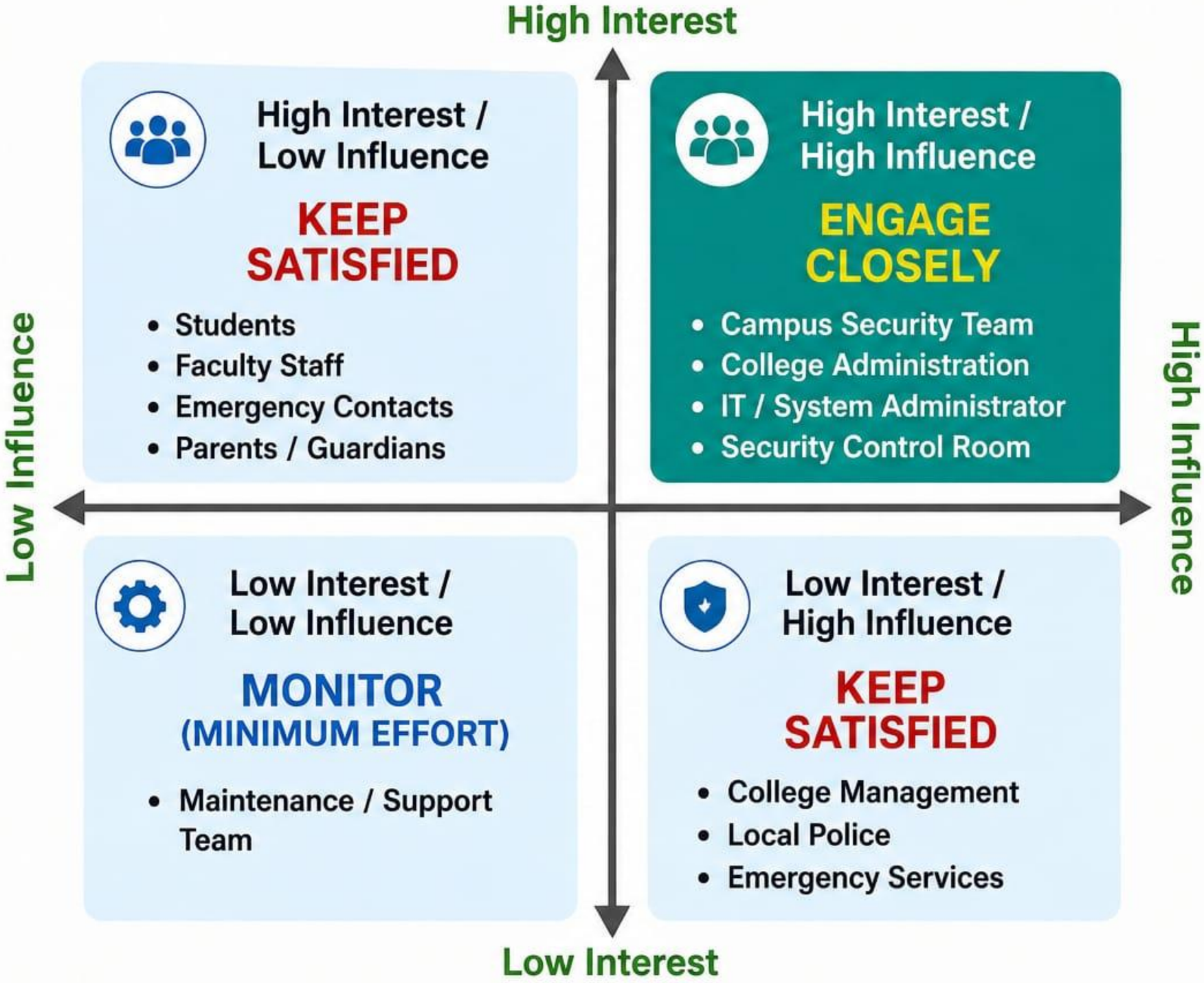
T10 : Stakeholder Links & Relationship Mapping

STAKEHOLDER LINKS & RELATIONSHIP MAPPING TEMPLATE

(Define the links and relationships between stakeholders)



T11 : Stakeholder Priority Mapping Matrix



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T12 : Stakeholder Analysis & Engagement Strategy

Stakeholder Analysis & Engagement Strategy

Key Stakeholders	Relationships	Stakeholder's Interest(s) in the Design Challenge	Impact Assessment	Strategies to Gain Support or Reduce Obstacles
Students	Use the system	Quick emergency help and safety	High	Simple SOS and easy-to-use app
Faculty Staff	Support students	Student safety and emergency alerts	High	Provide alerts and safety updates
Campus Security Team	Respond to incidents	Fast response and live monitoring	High	Security dashboard and real-time alerts
College Administration	Manage campus safety	Effective emergency management	High	Reports and centralized monitoring
IT / System Administrator	Maintain the system	System reliability and security	High	Regular maintenance and secure access
College Management	Oversee implementation	Improved campus safety	Medium	Performance reports and system outcomes
Local Police	External emergency support	Coordination during serious incidents	Medium	Share necessary incident information
Emergency Services	Provide emergency assistance	Quick emergency response	High	Direct emergency communication
Emergency Contacts	Receive emergency alerts	Student safety during emergencies	Medium	Instant notifications and updates
Parents / Guardians	Support students	Student safety and emergency awareness	Medium	Timely emergency notifications
Security Control Room	Monitor and coordinate	Real-time incident management	High	Live alerts and incident tracking
Maintenance / Support Team	Technical support	System availability and maintenance	Medium	Regular system checks and support

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T13 : Project Brief & Opportunity Framing

PROJECT BRIEF AND OPPORTUNITY FRAMING TEMPLATE

A. CHALLENGE OR PROBLEM DEFINITION		
Project Sponsor	**Organization Name**	Kongu Engineering College
	Address and Contact	Perundurai, Erode – 638 052, Tamil Nadu, India Phone: +91 4294 226500 | Email: info@kongu.ac.in
	Contact Person(s)	Dean – Student Affairs Phone: +91 4294 226577 | Email: dean_sa@kongu.ac.in
Project Title	Campus Shield – Smart Campus Safety & Emergency Response System	
Design Challenge	How might we design a smart, fast and reliable campus safety system that enables one-tap SOS alerts, real-time location sharing and quick emergency response to ensure the safety and security of students, faculty and staff on campus?	
Design Challenge Context and Background Info	• Emergencies on campus require immediate attention and quick response. • Delayed communication and lack of real-time location lead to slower response. • There is a need for an integrated digital system for alerts, tracking and monitoring. • This system will improve safety, reduce response time and maintain incident records.	
	Why does this Design Challenge matter to the organization • Ensuring the safety and well-being of the campus community is the top priority. A reliable emergency response system builds trust, enhances security and strengthens the institution's commitment to student welfare.	
Goals and objectives of the Design Challenge	• Provide one-tap SOS emergency support • Enable GPS-based location sharing • Provide real-time security monitoring • Improve quick emergency response • Maintain incident records and response status	
Target users of the Design Challenge	• Students • Faculty Staff • Campus Security Team	
Other key stakeholders	• College Administration • IT / System Administrator • College Management • Local Police	• Emergency Services • Parents / Guardians • Emergency Contacts • Security Control Room • Maintenance / Support Team
Previous efforts to solve this Design Challenge	• Manual emergency calls • Security patrols • Emergency contact systems	• Direct communication with security • Traditonal Traditona CCTV monitoring
Current activity system and process		

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T14 : Project Brief & Reframing Project Challenges

B. OPPORTUNITY FRAMING	
Real issues behing this Design Challenge	The main issues are slow emergency response, delayed communication, lack of real-time location, network failures, false alerts, and cybersecurity risks. These problems can delay assistance and affect student safety and data privacy.
Inspirations from others in solving this Design Challenge	The design is inspired by mobile SOS systems, GPS location sharing, security dashboards, emergency notifications, CCTV monitoring, and secure authentication systems used to improve emergency response and safety.
Teams contributions	The team contributed through stakeholder analysis, STEEP analysis, activity system mapping, emergency response and GPS ideas, security planning, project framing, documentation, and presentation.
Success criteria	The system should provide fast and reliable SOS alerts, accurate GPS location sharing, instant security notifications, real-time incident monitoring, reduced response time, secure student data, easy usability, and fewer false alerts.
"HOW MIGHT WE" Opportunity/ possibility statement	How might we create a fast, reliable and secure campus safety application that enables students to send one-tap SOS alerts, share their real-time location and receive quick emergency assistance?

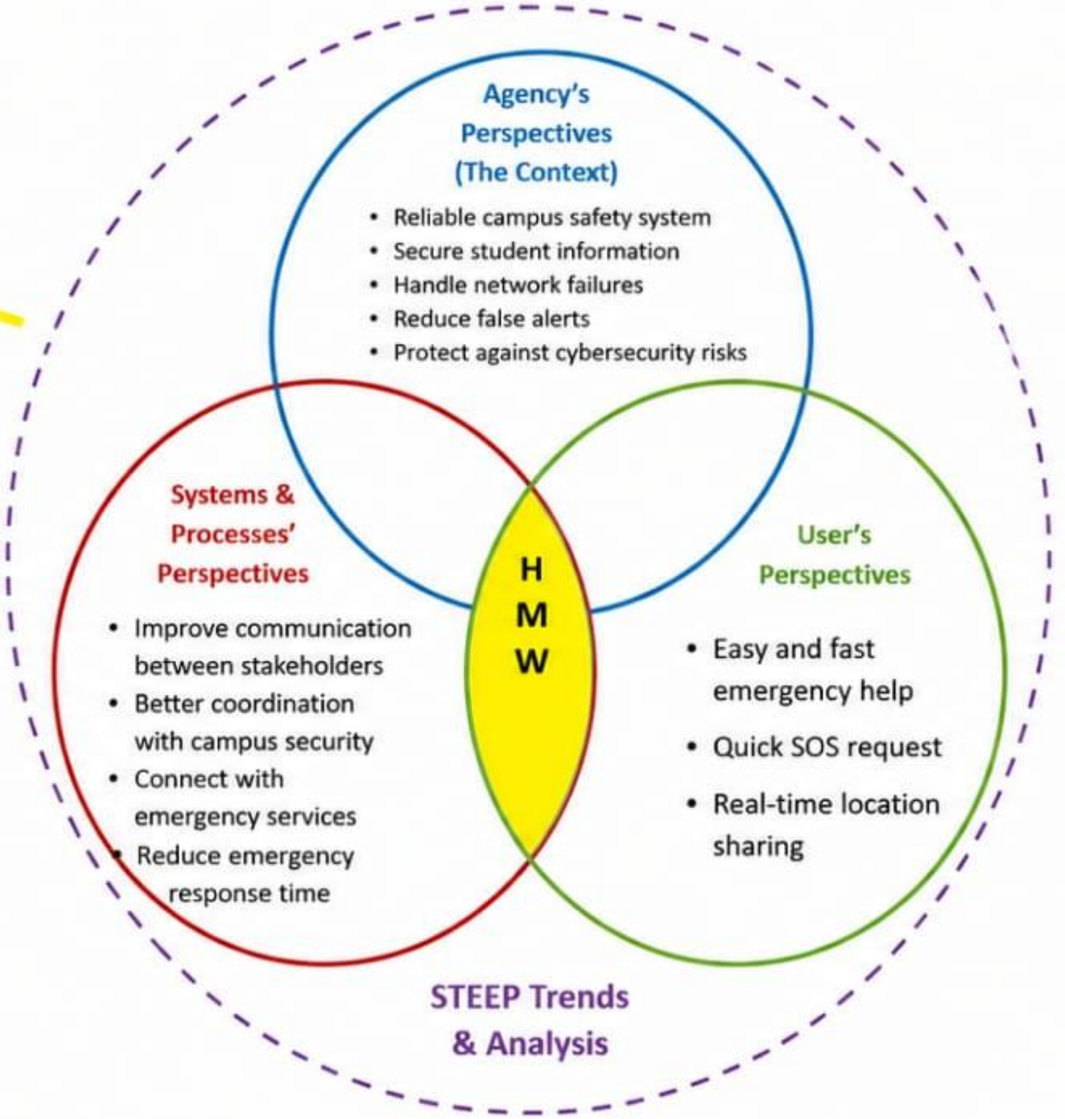
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T15 : Reframing the Opportunities

Reframe your opportunities and filter one

"How might we.....?"

1. How might we make emergency help easy and fast for students?
2. How might we improve coordination to reduce emergency response time?
3. How might we build a reliable and secure campus safety system?



SELECTED OPPORTUNITY

How might we design a simple, reliable and secure campus safety system that enables fast SOS alerts, real-time location sharing and quick emergency assistance?